

RESEARCH ARTICLE

Enhancing Multimodal Recommender System Performance Using Large Language Models

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ABSTRACT With the rapid advancement of e-commerce platforms and the overwhelming amount of information available online, recommender systems have emerged as essential tools for helping users efficiently discover items aligned with their preferences. Online reviews, in particular, offer rich information for understanding both user preferences and item features, and have been widely utilized in existing multimodal recommender systems. However, most prior approaches have not sufficiently addressed modality-specific noise in multimodal reviews, thereby constraining their capacity to precisely represent users' experiences. Furthermore, by merely extracting and fusing features from texts and images without considering the structural heterogeneity between modalities, these approaches often fall short in accurately predicting user preferences. To address these limitations, this study proposes multimodal aspect summarization for recommender systems (MAS-Rec), a novel model that leverages large language models (LLMs) to summarize multimodal reviews by aspect, thereby enhancing preference prediction. The proposed model is structured as follows: First, LLMs are employed to independently summarize user preferences and item features into aspects from textual and visual data, respectively. Second, these aspect-based summarizations are then encoded using a bidirectional encoder representations from transformers (BERT) to extract semantic features. Finally, a co-attention mechanism is applied to model the semantic interactions between user preferences and item features. Experimental results on two review datasets provided by Amazon.com demonstrate that MAS-Rec outperforms existing unimodal and multimodal models. These findings demonstrate that summarizing images by aspect using the LLM effectively captures the semantic connections between texts and images, thereby enabling more accurate prediction of user preferences compared to traditional approaches.

INDEX TERMS Recommender systems, large language models, modality heterogeneity, aspect-based summarization, co-attention mechanism.

I. INTRODUCTION

Advancements in information and communication technology (ICT) and e-commerce have facilitated convenient access to diverse items and services for users. However, the rapid expansion of online item availability has increased the difficulty of selecting items suited to individual user preferences [1]. In this context, recommender systems have gained attention as effective tools for reducing the time, cost, and

uncertainty associated with user decision-making [2], [3]. Early recommender systems adopted matrix factorization (MF) techniques that predict user preferences from historical purchase or rating data [4]. However, MF relies solely on ratings, which limits its ability to capture users' detailed preferences and underlying motivations [5]. Consequently, extensive research has explored the use of review texts to enhance recommendation models.

Review texts contain rich contextual information, including users' reasons for purchasing an item and their emotional attitudes toward it, thereby offering a more precise reflection

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of individual preferences [6]. For instance, Zheng et al. [7] applied convolutional neural networks (CNNs) to review texts in order to represent latent features of users and items, effectively capturing the interactions between them. Furthermore, Liu et al. [8] pointed out the limitations of previous studies in fully incorporating contextual information into preference prediction, and proposed a method that integrates features extracted from review texts to address this issue. In addition, Lim et al. [9] leveraged rich contextual information embedded in review texts by employing features extracted through bidirectional encoder representations from transformers (BERT) and the robustly optimized BERT pretraining approach (RoBERTa), thereby enhancing the representation of user and item features.

Images attached by users in reviews provide vivid and realistic information that is difficult to express through texts alone. This additional visual input enables the identification of item features that are not explicitly stated in the textual content [10]. Consequently, there has been growing interest in recommender systems that leverage multimodal reviews [11]. For example, He and McAuley [12] incorporated visual information into preference prediction by extracting visual features from images using the pretrained CNNs and integrating them into MF. To accurately capture modality-specific features essential for item recommendation, Liu et al. [13] employed distance covariance to learn independent features for each modality. Jeong et al. [14] extracted multimodal features based on RoBERTa and VGG16, and applied a co-attention mechanism to effectively reflect the complementary characteristics between texts and images.

These approaches exhibit structural limitations that hinder the effective incorporation of multimodal features in preference prediction. Review texts often include contextual noise and irrelevant content beyond item-related information [15]. Since prior studies on text modality typically utilize the entire review, they struggle to extract concise and relevant signals necessary for accurate preference prediction [16]. Review images are frequently captured by non-expert users, leading to the inclusion of noise and an overall degradation in image quality [17]. Consequently, the image modality struggles to contribute meaningful information to preference prediction based on extracted visual features [18]. Furthermore, as text is expressed in a symbolic form and images are represented as signal-based inputs, the two modalities differ in representational structure. Thus, naive fusion of these modalities often fails to establish a meaningful semantic alignment between them [19], [20].

Most existing multimodal recommender systems attempt to address this limitation by extracting modality-specific features and combining them through fusion mechanisms. However, such approaches often rely on fusion without sufficiently considering modality heterogeneity, which limits their ability to effectively capture the semantic relationships between modalities. To address this limitation, we propose a different approach that focuses on semantic alignment before feature extraction. Specifically, we leverage LLMs to convert

both textual and visual modalities into aspect-based textual representations, thereby enabling a unified representation of multimodal data. Based on this, co-attention is applied at a later stage to model semantic interactions between user preferences and item features.

Recently, large language models (LLMs) have garnered increasing attention as powerful tools for analyzing multimodal reviews. Trained on massive volumes of textual data, LLMs exhibit strong capabilities in contextual understanding and reasoning [21], [22]. Moreover, by incorporating vision encoders, LLMs are able to bridge the semantic gap between texts and images, thereby enabling a more accurate representation of item characteristics [23], [24]. Despite these advantages, LLMs remain underutilized in the domain of multimodal recommendation. In particular, approaches that summarize multimodal reviews by aspects using LLMs and subsequently employ these summaries as inputs to recommendation models remain underexplored. Aspect-based summarization offers a promising means to integrate the heterogeneous modalities into a unified textual representation, thereby facilitating fine-grained semantic alignment while richly capturing user experiences related to specific item aspects [25], [26].

Accordingly, this study proposes multimodal aspect summarization for recommender systems (MAS-Rec), a novel recommendation model that leverages LLMs to perform aspect-based summarization of review images and utilizes the resulting summaries to accurately predict user preferences. MAS-Rec effectively captures the semantic alignment between modalities while precisely reflecting users' emotions and opinions expressed in both texts and images for preference prediction. Specifically, MAS-Rec employs LLMs to perform aspect-based summarization on multimodal reviews, thereby extracting user preferences and item characteristics. These modality-specific textual summaries are then processed using a fine-tuned BERT to extract contextual features. Finally, a co-attention mechanism is applied to model the semantic interactions between users and items. The main contributions of this study are as follows:

- This study utilizes LLMs to perform aspect-based summarization of review images, thereby converting visual information into a unified representational format consistent with review texts. Based on this approach, we propose MAS-Rec, which effectively leverages the complementary characteristics of textual and visual modalities.
- This study applies a co-attention mechanism to aspect-based summarization generated by LLMs to predict user preferences. Whereas previous studies have utilized linguistic and visual features to capture overall user preferences, the proposed approach enables a more fine-grained understanding by accurately identifying users' detailed evaluations for each item aspect.
- This study evaluates recommendation accuracy using a review dataset from Amazon.com and conducts extensive experiments on aspect-based summarization

generated by LLMs. Experimental results demonstrate that aspect-based summarization improves recommendation accuracy compared to conventional methods that extract features separately from each modality.

The remainder of this paper is organized as follows. Section II reviews related work on existing multimodal recommender systems. Section III describes the framework of the proposed MAS-Rec model. Section IV presents the dataset, experimental setup, evaluation metrics, and baseline models. Section V reports the experimental results and provides analysis. Finally, Section VI summarizes the study, discusses its limitations, and outlines directions for future research.

II. RELATED WORK

A. REVIEW-BASED RECOMMENDER SYSTEM

With the advancement of ICT and e-commerce platforms, the volume of item information available online has grown exponentially, making it increasingly difficult for users to select items that match their preferences [27]. In response, recommender systems that analyze users' purchase histories to provide personalized suggestions have attracted significant attention [3], [28]. Early approaches primarily relied on MF, which depends on rating data alone, limiting its ability to reflect users' detailed experiences with items [4], [29]. To address this limitation, recent studies have actively explored the use of review texts, which offer rich insights into users' specific experiences [6], [30].

For example, Zheng et al. [7] applied convolutional neural networks (CNNs) to sets of user and item review texts to extract features, and subsequently incorporated the interactions between them into preference prediction. Liu et al. [8] noted that existing studies often fail to fully capture the rich information embedded in review texts, and thus proposed a multi-embedding approach. Lim et al. [9] leveraged BERT and RoBERTa to encode review text sets and combined the resulting representations to more effectively capture the rich information embedded in reviews for recommendation tasks. Furthermore, Kim et al. [31] emphasized the necessity of considering both local and global information to more accurately reflect meaningful content in review texts within recommendation systems. To this end, local features were extracted using CNNs, while global information was obtained via RoBERTa; these were then fused and utilized for user preference prediction. While these studies successfully incorporate users' concrete experiences from review texts, they share a common limitation. Specifically, they do not explicitly identify aspects, which introduce contextual noise unrelated to the actual item features [16].

Aspects refer to fine-grained evaluation targets through which users express their emotions or opinions regarding various elements of an item [32], [33]. Accordingly, users' experiences are represented at the aspect level of items and play a crucial role in preference evaluation [34]. Moreover, since individuals differ in the aspects they consider important depending on their personal preferences, it is necessary to

account for the varying influence of each aspect on user preferences [35]. Accordingly, aspect-based recommender systems that extract and model aspects from review texts have been actively studied. Chin et al. [36] noted that conventional recommender systems fail to identify aspects that are critical for preference prediction within review texts. To address this issue, they employed CNNs and an attention mechanism to extract salient aspects from reviews, and applied a co-attention mechanism to model user-item interactions at the aspect level. Extending this line of work, Li and Xu [37] improved aspect extraction by jointly modeling local and global contexts using CNNs and bidirectional long short term memory (Bi-LSTM), addressing the limitations of earlier methods that focused solely on local features. These findings suggest that aspects constitute essential semantic units that shape the overall meaning of reviews and allow for a clearer representation of user preferences [38].

However, most existing studies utilize the entire review texts, which often includes information not directly related to item aspects, thereby introducing noise during the feature extraction process [15]. In addition, although aspect extraction methods based on pretrained models such as BERT have been actively explored, they typically require domain-specific datasets and complex pretraining procedures, which may limit their applicability to real-world domains [33], [39]. To address these limitations, this study proposes MAS-Rec, a recommendation model that integrates aspect-based summarization of multimodal reviews generated by LLMs. By leveraging their contextual understanding and reasoning capabilities, LLMs can effectively extract information related to item aspects from both texts and images [40]. Moreover, unlike techniques employed in prior studies, the application of instruction prompts to LLMs enables the selective representation of aspect-specific information that is critical for preference prediction. As a result, the proposed MAS-Rec accurately captures the target aspects of item related opinions while incorporating the specific context of user experiences into preference prediction.

B. MULTI-MODAL RECOMMENDER SYSTEM

Recently, with the advancement of information and communication technologies and the expansion of online platforms, users have increasingly utilized both texts and images to share their experiences with purchased items [14]. In particular, review images depict how items are used in real-world contexts, thereby helping to better understand how users interact with the items [41]. As visual representations of actual user experiences, review images can provide valuable information for predicting user preferences. Against this background, there has been growing interest in incorporating multimodal information into recommender systems [11], [42].

For example, He and McAuley [12] proposed a multimodal recommender system that integrates the visual features of items into a MF framework to examine the impact of images on user preference prediction. Their results demonstrated that

visual information complements rating data and contributes to more accurate preference prediction. In another study, Chen et al. [43] aimed to improve image content recommendations for Twitter users by extracting texts embedded in images using optical character recognition (OCR) in combination with hashtags, and incorporating this information into an MF-based model. This approach effectively utilized contextual information present in images for content recommendation. These studies highlight the value and potential of leveraging multimodal information in user preference prediction.

However, early studies on multimodal recommender systems often overlooked the complementary characteristics of text and image modalities. While text conveys users' subjective opinions and emotional expressions, images provide the visual aspects of items; thus, the two modalities can effectively complement each other [10], [42]. To take advantage of this, more recent research has focused on developing multimodal recommender systems that explicitly consider the complementary nature of different modalities.

Liu et al. [13] pointed out that prior studies treated each modality equally without considering the relative importance of information, and failed to disentangle item characteristics at the aspect level. To address this issue, they proposed a method that independently learns modality-specific features by factor, while incorporating an attention mechanism to capture salient aspects. Similarly, Jeong et al. [14] predicted user preferences more accurately by incorporating the complementary characteristics of multimodal information. Specifically, they extracted textual and visual features using RoBERTa and VGG16, respectively, and applied a co-attention mechanism to selectively emphasize information critical for preference prediction. Furthermore, Wang et al. [44] applied a co-attention mechanism to multimodal features in order to selectively extract important information and subsequently integrated the multimodal representations with user latent factors.

Although these studies effectively incorporate the complementary characteristics across modalities into recommendation systems, they face limitations in adequately filtering the noise present in review images. Review images are typically captured by non-expert users and are therefore subject to various environmental factors such as lighting conditions, exposure, focus, and lens quality, all of which can introduce noise into the visual content [17]. Such noise hinders conventional recommendation systems from effectively capturing meaningful item related information from images [18]. In addition, existing studies primarily rely on fusion-based approaches to extract and combine multimodal features, without sufficiently addressing modality heterogeneity [45], [46]. Modality heterogeneity refers to the differences in representational formats across modalities, with texts and images exhibiting particularly high levels of heterogeneity [47]. Such differences in representational form hinder the semantic alignment between texts and images, thereby limiting

the ability to capture user preferences with fine granularity through simple fusion of the two modalities [19].

To address this limitation, this study proposes MAS-Rec, a recommendation system that leverages LLMs to convert images into texts for user preference prediction. MAS-Rec incorporates LLMs, which utilize a contrastively pretrained vision encoder to transform review images into contextual information, thereby mitigating modality heterogeneity [23]. In addition, this study adopts an aspect-based summarization approach for images, enabling richer descriptions of key item aspects and detailed information. Through this process, MAS-Rec effectively utilizes the complementary characteristics of multimodal data to enhance the accuracy of preference prediction.

C. LARGE LANGUAGE MODELS

As user interactions on online platforms continue to increase, the volume of textual data available online has grown rapidly [42]. To efficiently process such large-scale textual data, pretrained LLMs have gained significant attention in the field of NLP. Models such as GPT-4 and LLaMA-3 are pretrained on massive text corpora and demonstrate strong capabilities in contextual understanding and reasoning [21], [22]. Moreover, LLMs can perform a wide range of tasks without additional fine-tuning through the provision of appropriate instructions, a feature known as in-context learning [48]. As a result, LLMs have shown impressive performance across various tasks, including machine translation [49], text summarization [40], and semantic inference [50].

The advancement of LLMs has also accelerated the development of multimodal models capable of integrating and analyzing diverse types of data, such as texts and images. Recently, multimodal models have evolved into architectures that combine a single vision model with an LLMs, enabling richer understanding of multimodal information compared to traditional approaches [24]. In particular, Liu et al. [23] proposed an LLM capable of converting multimodal inputs into coherent textual outputs for tasks such as image captioning, inference, and dialogue. This approach has shown strong potential for addressing modality heterogeneity. Specifically, the model extracts visual information using a vision transformer (ViT) and incorporates it into the input of the LLMs, allowing for effective modeling of semantic interactions between texts and images.

Therefore, this study proposes MAS-Rec, a recommendation system that leverages LLMs to capture semantic interactions across modalities while precisely expressing user opinions on item aspects. Using LLMs to summarize images by aspect allows for a more fine-grained depiction of the contextual meaning conveyed by images, thereby improving upon traditional approaches that rely solely on visual features [51]. Moreover, the system enables explicit identification of item aspects that users value most, thereby incorporating critical information into preference prediction.

III. METHODOLOGY

A. MAS-Rec FRAMEWORK

Recommender systems aim to promote user purchases by filtering the vast amount of online information and recommending items that align with individual preferences. Traditional recommender system research has primarily focused on predicting user preferences based on review texts, which contain detailed user experiences with items. In contrast, images can complement textual information by depicting how items are used in real-world contexts [41]. In response, multimodal recommender systems that incorporate both textual and visual information have attracted increasing research attention. However, most existing studies do not explicitly account for the heterogeneous structures and representations of texts and images. Instead, they typically extract features from each modality independently and combine them at a later stage.

To address this limitation, we propose MAS-Rec, a multimodal recommendation model that utilizes LLMs to perform aspect-based summarization of review images. This approach enables a more fine-grained understanding of semantic interactions across modalities. Specifically, our approach converts both textual and visual information into unified aspect-based textual representations, enabling semantic alignment between modalities prior to feature extraction. This design helps mitigate the impact of modality heterogeneity by enabling the model to focus on semantically relevant information. As a result, the model can more effectively capture fine-grained semantic interactions across modalities.

The architecture of the proposed MAS-Rec is illustrated in Figure 1 and consists of two main components: the aspect-based summarization module and the rating prediction module. First, in the aspect-based summarization module, review texts and images are processed using meta-llama-3-8b-instruct (LLaMA)¹ and llava-1.6-mistral-7b (LLaVA)², respectively, to generate textual summaries. To enable LLaVA to more accurately interpret the item shown in the images, the item title is provided as an additional input alongside the images. Second, the rating prediction module predicts user preferences by utilizing aspect-based summarization generated by the LLMs. The textual and visual summaries are transformed into contextual representations of users and items using a fine-tuned BERT and a multi-layer perceptron (MLP), respectively. These representations are then combined with interaction features between users and items to capture latent preference patterns. Finally, a co-attention mechanism is applied to learn the semantic interactions between the two modalities, which are used to estimate user preferences.

B. ASPECT-BASED SUMMARIZATION MODULE

The aspect-based summarization module aims to generate key features essential for preference prediction by summarizing review texts and images at the aspect level. For the textual modality, LLaMA is utilized to capture users' preferences toward each item aspect with fine granularity.

¹<https://huggingface.co/meta-llama/Meta-Llama-3-8B-Instruct>

²<https://huggingface.co/liuhaotian/llava-v1.6-mistral-7b>

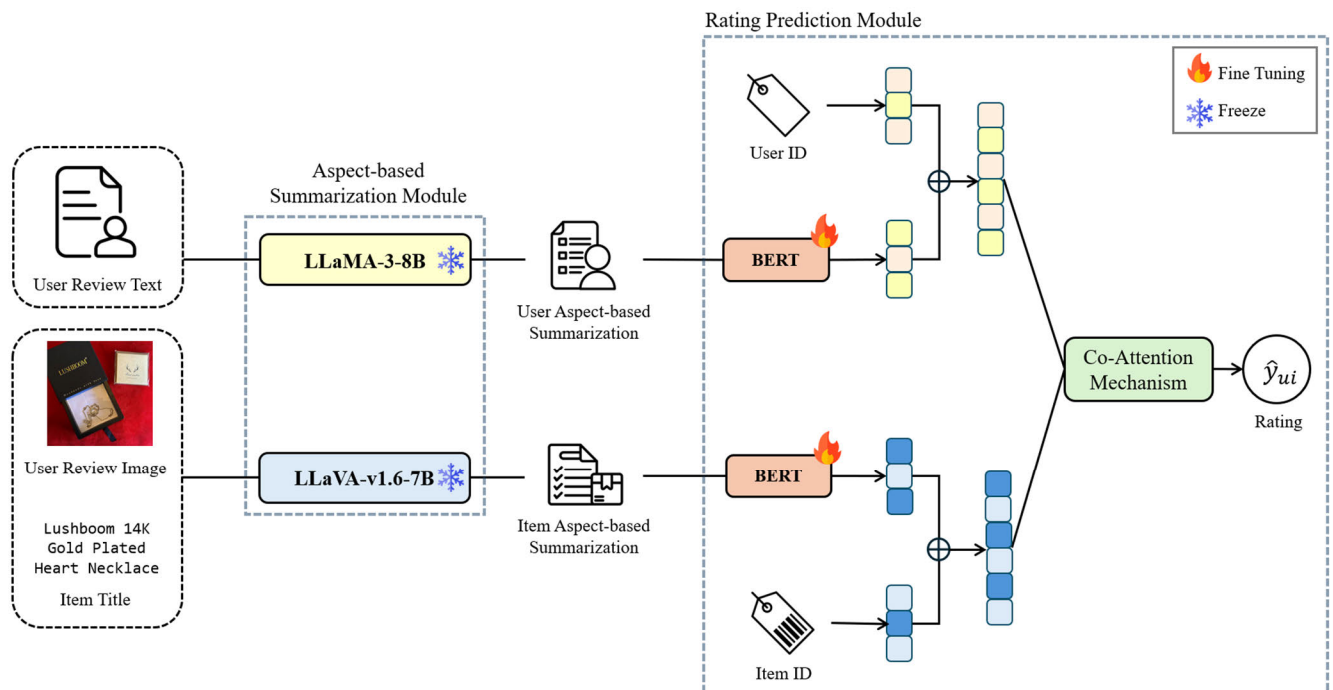


FIGURE 1. The overall framework of the MAS-Rec model.

LLaMA is an LLM released by Meta AI, which has been instruction-tuned to effectively handle various tasks based on specific prompts. In this study, LLaMA is applied to review texts to perform aspect-based summarization. To facilitate more effective analysis, instruction-based prompts were incorporated into the input of LLaMA, as illustrated in TABLE 1. These prompts were designed to enable the model to fully leverage its prior knowledge and reasoning capabilities, thereby producing more accurate representations of user preferences [48]. Additionally, the prompts were carefully designed to reflect the domain-specific characteristics of e-commerce platforms such as Amazon.

TABLE 1. Instruction prompt template for aspect-based summarization using LLaMA.

Role	You are an assistant summarizing user preferences from user reviews in the domain of Clothing Shoes and Jewelry.
Task	Your task is to generate natural-sounding, multi-sentence summaries that reflect user preferences based on various product aspects.
Aspect Categories	Predefined aspect categories: Styles, Colors, Fits, Patterns, Brands. If you identify other relevant aspects not covered by these categories, you may include them.
Input	<Review Text>

As shown in Table 1, the instruction prompt, excluding the input, is composed of three components: Role, Task, and Aspect Categories. First, the Role component specifies the model's role and the domain of the review texts to be analyzed, enabling the LLMs to effectively leverage its domain knowledge and reasoning capabilities. Second, the Task component instructs the model to summarize user preferences for an item at the aspect level in the form of naturally articulated sentences, thereby richly capturing users' concrete experiences with the item. Third, the Aspect Categories component defines key aspects in advance, based on prior studies, to facilitate aspect-based summarization of user preferences. For example, in the fashion domain, the categories include Styles, Colors, Fits, Patterns, and Brands [52], [53], while in the mobile phone domain, the categories consist of Features, Qualities, Designs, Durabilities, and Brands [54], [55]. In addition, the prompt is designed to allow the inclusion of additional meaningful aspects that may emerge from review texts beyond the predefined categories.

For visual information, LLaVA is employed to effectively represent item features by utilizing review images and item titles. This model consists of a pretrained vision encoder and an LLM. It is designed to extract visual information and summarize it into natural language for various image-based tasks such as question answering, description generation, and visual reasoning [23]. In addition, it is instruction-tuned, allowing it to flexibly handle diverse types of visual inputs. Based on this capability, visual information can be converted

into a unified representational format consistent with review texts, enabling the model to effectively reflect the complementary characteristics of the two modalities [56]. Therefore, in this study, LLaVA is applied to review images and item titles to perform aspect-based summarization. Similar to LLaMA, instruction prompts are added to support effective analysis of item characteristics represented in review images. These prompts are presented in TABLE 2.

TABLE 2. Instruction prompt template for aspect-based summarization using LLaVA.

Task	Product title: <Product Title> Analyze this fashion product image and its title. Summarize the product's features from this image in the Clothing Shoes and Jewelry domain.
Aspect Categories	Focus on product aspects such as Styles, Colors, Fits, Patterns, Brands. You may include other relevant aspects if mentioned.
Input	<Review Image>

As shown in Table 2, the instruction prompt, excluding the input, is concisely composed of two components: Task and Aspect Categories, with the Role component used in the LLaMA prompt omitted. This design considers that LLaVA is primarily trained on visual information. In addition, the item title is included to ensure that LLaVA can clearly identify the target item for analysis. Consequently, the final aspect-based summarizations for texts and images are denoted as S_{text} and S_{image} respectively.

C. RATING PREDICTION MODULE

1) USER FEATURE EXTRACTOR

The user feature extractor is designed to represent user characteristics by integrating the textual aspect-based summarization S_{text} generated by LLaMA. First, context-aware embeddings are extracted through aspect-based summarization of the user's textual information. To effectively reflect key features that are crucial for preference prediction in the recommendation task, BERT is fine-tuned and employed for this purpose [48]. BERT analyzes sentence meaning bidirectionally, effectively capturing the flow of information and context within the sentence. Since the final [CLS] token of BERT represents the overall meaning of the texts, it is used as the feature vector in this study. Accordingly, the user text features derived through aspect-based summarization are defined as follows:

$$h_{text} = BERT(S_{text}), \quad (1)$$

where h_{text} denotes a 768-dimensional feature vector derived through aspect-based summarization of the user's textual information. To enhance the semantic representation of the user text features, a two-layer MLP is applied to reduce the item text features to 256 dimensions. Additionally, the rectified linear unit (ReLU) activation function, which is

commonly used in deep learning, is employed to effectively capture nonlinear interactions within the user text features. As a result, the contextual user representation reflecting user preferences $\tilde{h}_{text}$ is obtained.

At the same time, the user ID is transformed into a dense vector representation. This allows the model to effectively capture the user's latent preferences by identifying sparse interactions between users and items. This process is implemented by applying one-hot encoding followed by an embedding technique to the user ID, and is defined as follows:

$$p_u = P^T v_u^U, \quad (2)$$

where $P \in \mathbb{R}^{m \times d}$ denotes the embedding matrix used to embed user IDs, m represents the total number of unique users, and v_u^U is the one-hot vector corresponding to user u . The resulting user latent vector p_u is optimized during training. Subsequently, to minimize information loss, the extracted user contextual feature $\tilde{h}_{text}$ is concatenated with the user latent vector p_u . This process is defined as follows:

$$V_{user} = [\tilde{h}_{text}; p_u], \quad (3)$$

where $[\cdot]$ denotes the vector concatenation operation. V_{user} represents the user feature, which is subsequently fed into the semantic interaction network to model the semantic alignment across modalities.

2) ITEM FEATURE EXTRACTOR

The item feature extractor aims to represent item features by combining aspect-based summarization S_{image} generated by LLaVA with the item ID. First, a fine-tuned BERT is applied to the aspect-based summarization to extract context-aware embeddings. Similar to the user text features, the final [CLS] token of BERT is used as the feature vector to effectively capture item-specific characteristics. Accordingly, the item text features are represented as follows:

$$h_{image} = BERT(S_{image}), \quad (4)$$

where h_{image} denotes a 768-dimensional feature vector derived through aspect-based summarization of the item's textual information. Subsequently, a two-layer MLP with ReLU activation is applied to reduce the item text features to 256 dimensions. As a result, the contextual item representation reflecting item features $\tilde{h}_{image}$ is obtained.

In parallel, item IDs are first subjected to one-hot encoding and subsequently transformed into dense vectors through embedding techniques. This process enables the extraction of key interactions between users and items, thereby representing the latent factors of the items. The procedure is defined as follows:

$$q_i = Q^T v_i^I, \quad (5)$$

where $Q \in \mathbb{R}^{n \times d}$ denotes the embedding matrix applied to item IDs. n represents the number of unique items and v_i^I refers to the sparse vector corresponding to item. The vector q_i denotes the latent vector of item i , which represents

the item's latent preferences based on historical interactions. Subsequently, the extracted contextual feature of the item $\tilde{h}_{image}$ is combined with the item's latent vector q_i . This process is formally defined as follows:

$$V_{item} = [h_{image}; q_i] \quad (6)$$

where V_{item} represents the item features. This vector is subsequently fed into the semantic interaction network, where it is utilized to model the semantic interactions across different modalities.

3) SEMANTIC INTERACTION NETWORK

The semantic interaction network focuses on modeling the complementary characteristics across different modalities. In this study, a co-attention mechanism is applied to both user and item features to capture semantic interactions between modalities. As the item features are derived through aspect-based summarization and transformed into contextual representations, textual and visual modalities are aligned within a unified representation space, enabling the effective integration of complementary information.

Inspired by previous studies, this work implements the co-attention mechanism based on the multi-head attention architecture of the Transformer [14], [44]. This design allows each attention head to capture distinct aspects of semantic interactions. As a result, the recommender system can attend to informative features from multiple perspectives, leading to improved preference prediction [15]. The self-attention mechanism is formally defined as follows:

$$Attention(Q, K, V) = softmax\left(\frac{QK^T}{\sqrt{d_k}}\right)V, \quad (7)$$

where Q , K and V represent the query, key, and value matrices, respectively, which are used to compute attention weights. d_k denotes the dimensionality of the query and key vectors. In this process, the relevance scores are computed via the dot product between queries and keys, and then scaled by $\sqrt{d_k}$ to prevent the values from becoming excessively large [57]. This scaling enables the model to focus on key features within the input sequence that are critical for preference prediction.

Furthermore, multi-head attention employs multiple attention heads in parallel, allowing the model to capture diverse types of information and enhance the richness of the learned representations. Specifically, Q , K and V is projected into an independent linear subspace, facilitating the effective learning of various semantic interactions. The formulation is presented in Equation (8).

$$MultiHead(Q, K, V) = [head_1; head_2; \dots; head_N] W^O,$$

where

$$head_i = Attention(QW_i^Q, KW_i^K, VW_i^V), \quad (8)$$

where W_i^Q , W_i^K and W_i^V denote the projection matrices for the i -th attention head, while W^O refers to the projection

matrix used to transform the final output back to the original input dimension. The index i indicates the specific attention head, and each head is trained independently without sharing weights.

Based on this architecture, the proposed MAS-Rec applies a co-attention mechanism to effectively capture semantic interactions between user and item features. This mechanism enables mutual referencing between the two feature types, allowing the model to selectively attend to semantically relevant representations in the counterpart feature [58]. Moreover, it facilitates the exchange of semantically aligned information between the two modalities, leading to richer and more expressive feature representations [44]. In addition, since image data has previously been transformed into contextual features, the model is better able to capture accurate semantic interactions between modalities. Accordingly, the process of modeling the complementary relationship between user and item features via the semantic interaction network is defined as follows:

$$\begin{aligned}\tilde{V}_{user} &= \text{MultiHead}(V_{user}, V_{item}, V_{item}), \\ \tilde{V}_{item} &= \text{MultiHead}(V_{item}, V_{user}, V_{user}),\end{aligned}\quad (9)$$

where $\tilde{V}_{user}$ and $\tilde{V}_{item}$ represent the semantically integrated user and item features, respectively. These representations are refined by selectively incorporating elements that exhibit high semantic relevance to their counterpart features. As a result, the model produces more sophisticated and discriminative feature representations.

4) RATING PREDICTION NETWORK

The rating prediction module estimates user preferences by leveraging user and item feature vectors that incorporate semantic interactions across modalities. In this study, to effectively leverage the complementary characteristics of multimodal reviews, visual information is first transformed into contextual representations and then integrated into the preference prediction process. To achieve this, the semantically integrated user and item representations $\tilde{V}_{user}$ and $\tilde{V}_{item}$ are first obtained from the semantic interaction network. These vectors are then concatenated and passed through an MLP to predict user preferences. This process is formally defined as follows:

$$\begin{aligned}f_0 &= [\tilde{V}_{user}; \tilde{V}_{item}], \\ &\quad \dots \\ f_1 &= \sigma(W_1 f_0 + b_1), \\ \hat{y}_{u,i} &= W_L f_{L-1} + b_L,\end{aligned}\quad (10)$$

where f_0 represents the final feature vector obtained by combining user and item representations. W_L , b_L , and σ denote the weight matrix, bias term, and ReLU activation function of the L -th layer, respectively. In this study, user preference is quantified as a rating score, and the prediction task is formulated as a regression problem. Accordingly, $\hat{y}_{u,i}$ indicates the predicted rating that user u is expected to assign to item i .

The predicted score $\hat{y}_{u,i}$ is optimized to minimize the discrepancy with the actual rating, and the mean squared error (MSE) is adopted as the loss function for this purpose. The formulation is defined as follows:

$$L_{MSE} = \frac{1}{N} \sum_{i=1}^N (y_{u,i} - \hat{y}_{u,i})^2, \quad (11)$$

where $y_{u,i}$ denotes the ground-truth rating, and N represents the number of predicted ratings. During this process, the BERT encoders for each modality are fine-tuned jointly with the overall model to effectively capture features that are critical for preference prediction. The fine-tuning is integrated into the loss function optimization. Furthermore, to enhance training efficiency and stability, the adaptive moment estimation with weight decay (AdamW) optimizer is employed.

IV. EXPERIMENTS

A. DATASETS

To evaluate the performance of the proposed MAS-Rec model, we employed the Amazon dataset, which is widely used in recommender system research [36], [44]. This dataset spans various item categories and consists of user-generated reviews and ratings following item purchases on the e-commerce platform. In this study, we selected two representative categories: Clothing, Shoes and Jewelry, which represents experience goods, and Cell Phones and Accessories, which represents search goods. The selection was based on both the characteristics of the items and the potential applicability of visual information in recommendation scenarios.

Experience goods involve characteristics that are difficult to evaluate accurately without actual usage. In contrast, search goods allow users to assess features such as specifications, performance, and price more objectively prior to purchase. Consequently, user preferences for experience goods are more strongly influenced by individual experiences, while search goods tend to be shaped by more objective and rational evaluations. These two categories not only exhibit distinct consumer behavior patterns but also include ratings, textual reviews, and item images, which make them well suited for training multimodal recommender systems. Therefore, although only two subsets were used, they provide a representative and diverse evaluation setting for validating the effectiveness and generalizability of the proposed model across different recommendation scenarios. In addition, these categories exhibit different patterns of user preference formation and multimodal information utilization. In particular, visual information tends to play a more experiential role in fashion-related products, whereas objective aspect evaluation is more prominent in mobile phone products. Therefore, the selected datasets provide an effective experimental setting for evaluating whether the proposed framework can effectively capture semantic interactions across different recommendation scenarios.

To ensure effective experimentation, this study utilized only user reviews that included both textual and visual

information. Additionally, only users and items with at least three associated reviews were retained. To evaluate the performance of the proposed MAS-Rec model and the baseline methods, the entire dataset was randomly split into 70% for training, 10% for validation, and 20% for testing. The basic statistics of the final dataset are presented in TABLE 3.

TABLE 3. Summary of dataset characteristics.

Dataset	Number of Reviews	Number of Users	Number of Items	Sparsity
Clothing, Shoes and Jewelry	66,718	7,449	16,099	99.944%
Cell Phones and Accessories	23,491	4,149	4,975	99.886%

B. EVALUATION METRICS

To evaluate the performance of the proposed MAS-Rec model and the baseline methods, we employed commonly used evaluation metrics for rating prediction, including mean absolute error (MAE), MSE, root mean squared error (RMSE), mean absolute percentage error (MAPE). MAE measures the average absolute difference between predicted and actual ratings, reflecting errors in a linear manner. In contrast, MSE squares each error before averaging, assigning greater penalty to larger deviations. RMSE is computed by taking the square root of MSE, enabling error interpretation in the same numerical unit as the actual ratings. Finally, MAPE expresses errors in percentage form, providing an intuitive understanding of their relative magnitude. The formulations of MAE, MSE, RMSE, and MAPE are given below.

$$MAE = \frac{1}{N} \sum_{i=1}^N |y_{u,i} - \hat{y}_{u,i}|, \quad (12)$$

$$MSE = \frac{1}{N} \sum_{i=1}^N (y_{u,i} - \hat{y}_{u,i})^2, \quad (13)$$

$$RMSE = \sqrt{\frac{1}{N} \sum_{i=1}^N (y_{u,i} - \hat{y}_{u,i})^2}, \quad (14)$$

$$MAPE = \frac{1}{N} \sum_{i=1}^N \left| \frac{y_{u,i} - \hat{y}_{u,i}}{y_{u,i}} \right| \times 100. \quad (15)$$

where N denotes the number of predicted ratings, $y_{u,i}$ refers to the ground-truth rating that user u assigned to item i , and $\hat{y}_{u,i}$ indicates the rating predicted by the model. By incorporating all four evaluation metrics, it becomes possible to comprehensively assess prediction accuracy, sensitivity to outliers, and relative error, thereby enhancing the reliability of recommender system evaluation [59].

C. BASELINE MODELS

In this study, we propose a recommendation model that effectively captures the complementary characteristics of

multimodal reviews through aspect-based summarization of images. To validate the preference prediction accuracy of the proposed model, MAS-Rec, we conducted experiments comparing its performance with several baseline models. The selected baselines include NeuMF, which leverages user-item interactions; text-based recommender systems such as ANR, AFRAM, and MFNR; and multimodal recommender systems including CAMRec and MMCN.

- NeuMF [60]: A deep learning-based recommender system that combines Generalized Matrix Factorization with a MLP to learn both linear and nonlinear relationships between users and items. This model predicts user preferences based solely on user-item interactions.
- ANR [36]: A model that predicts user preferences by extracting aspects from sets of user and item review texts. It employs an attention mechanism and a CNN to extract aspects, and dynamically captures the relationships between user and item aspects through a co-attention mechanism.
- AFRAM [37]: A recommender system that extracts user and item aspects from review texts by leveraging both local and global textual features via LSTM and CNN. The model uses sets of reviews to enhance aspect extraction and applies an attention mechanism to estimate the importance of each aspect for preference prediction.
- MFNR [9]: A model that predicts user preferences by capturing contextual features from sets of user and item reviews. It utilizes BERT and RoBERTa to encode the review texts and combines them using a multi-embedding strategy.
- CAMRec [14]: A multimodal recommender system that leverages the complementary characteristics of different modalities for preference prediction. It extracts features from texts and images using pretrained RoBERTa and VGG16, and employs a co-attention mechanism to identify modality-specific features that are important for preference prediction.
- MMCN [44]: A bipartite graph-based recommender system that provides personalized recommendations using multimodal features. The model mitigates modality heterogeneity through a co-attention mechanism and hierarchical contrastive loss, effectively capturing key features relevant to user preferences.

D. IMPLEMENTATION DETAILS

The hyperparameters of the proposed MAS-Rec model were determined through a representative configuration-based search to ensure experimental stability. For BERT fine-tuning in the proposed model, the batch size was set to 32, the learning rate to $1e-5$, and the optimizer was configured as AdamW. The embedding dimension for both users and items was set to 64, and the number of multi-heads was set to 8.

For fair performance comparison, the baseline models were implemented using the hyperparameter settings reported in their original studies. If specific values were not provided, empirically optimal configurations were applied.

To ensure consistency across experiments, both CAMRec and MMCN, which are multimodal recommender systems, were configured to extract textual and visual features using RoBERTa and VGG16, respectively. In MMCN, item level multimodal representations were generated by averaging the feature vectors of all reviews associated with the same item. Furthermore, to improve the rating prediction performance of MMCN, we adopted MSE as the primary loss function and incorporated the hierarchical contrastive loss proposed in the original study as a weighted auxiliary objective.

All experiments were conducted for a maximum of 100 epochs, with early stopping applied if validation loss did not improve for five consecutive epochs. To ensure reproducibility, each model was trained and evaluated five times, and the average results were reported. All experiments were performed on the Google Colab platform using the Intel(R) Xeon(R) CPU and the NVIDIA A100 GPU.

The LLM-based aspect summarization was performed as an offline preprocessing step prior to model training. The processing time varies depending on the model and prompting strategy. For example, generating summaries for 10,000 samples required approximately five hours. Additional overhead was observed when using more complex prompts such as Reasoning. Despite this computational cost, the summarization process is executed only once and is decoupled from both the training and inference stages of the recommendation model. As a result, it does not affect the runtime efficiency of MAS-Rec. Furthermore, in practical deployment scenarios, this preprocessing cost can be mitigated through parallelization or batch processing, enabling the proposed approach to scale to larger datasets.

E. CASE STUDY ON ASPECT-BASED SUMMARIZATION WITH LLMs

In this study, the proposed MAS-Rec model employs LLMs to generate textual summaries of images, with the aim of more precisely incorporating the complementary characteristics across modalities into preference prediction. This approach preserves the information contained in images while unifying the representational format of texts and images, thereby enabling fine-grained analysis of semantic interactions between modalities [56]. In this section, we examine whether LLaVA can effectively represent item characteristics without information loss, based on review images and item titles. The aspect-based summarization results generated by LLaVA are presented in FIGURE 2.

The results confirm that LLaVA can effectively analyze item characteristics through aspect-based summarization. The item depicted in the review images is a men’s wristwatch featuring a stainless-steel band and a black analog movement, which together convey a visually refined and distinct impression. With respect to the Styles aspect, LLaVA assessed the item as suitable for both casual and business settings. For the Colors aspect, the model effectively described the visual characteristics of the watch using the expression “black face with white markings.” In addition, regarding the Fits aspect, it inferred that the fit of the watch is adjustable, while for the Patterns aspect, it described the design as simple. Finally, for the Brands aspect, the model accurately identified the brand name “Olika” as indicated in the image. These results demonstrate that LLMs are capable of richly representing item characteristics at the aspect level based on review images and item titles.

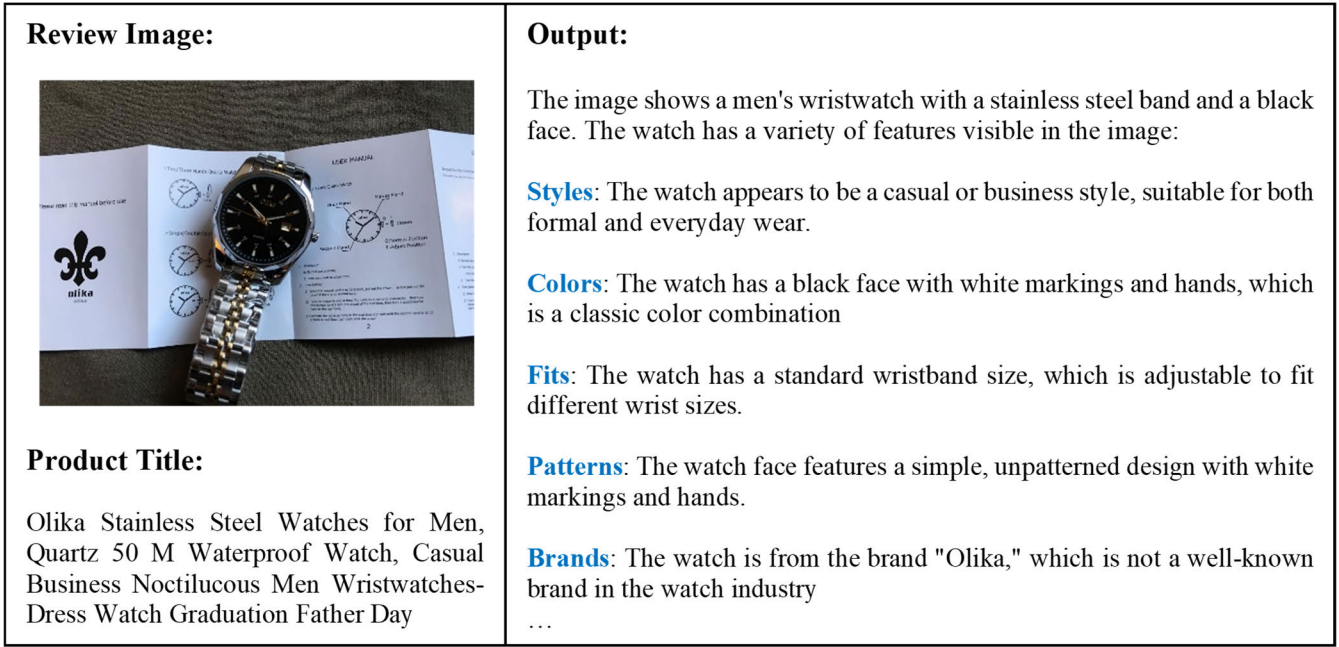


FIGURE 2. LLM-generated representations derived through aspect-based summarization of item images and titles.

V. EXPERIMENTAL RESULTS

A. PERFORMANCE COMPARISON WITH BASELINE MODELS

To validate the effectiveness of the proposed MAS-Rec model, we conducted extensive performance comparison experiments against various baseline models. The results demonstrate that MAS-Rec consistently outperforms all baseline models across all evaluation metrics. Detailed experimental results are presented in TABLE 4, from which the following conclusions can be drawn.

First, among the baseline models, NeuMF exhibited the lowest performance. This result suggests that relying solely on user-item interaction data is insufficient for accurately capturing user preferences. Second, ANR and AFRAM, which extract aspects from review texts, demonstrated superior performance compared to NeuMF, which depends exclusively on interaction data. These findings indicate that item aspects derived from review texts effectively reflect users' detailed experiences, thereby enhancing the accuracy of preference prediction. Notably, AFRAM achieved better performance by capturing both local and global contextual information from review texts, allowing the model to identify key aspects that are critical to user preferences. Third, MFNR, which leverages contextual features from review texts, outperformed aspect-based recommender systems such as ANR and AFRAM. This result implies that integrating contextual information extracted using pretrained language models, such as BERT and RoBERTa, is beneficial. Such integration enables a richer and more nuanced representation of user preferences expressed in review texts.

TABLE 4. Performance comparison of MAS-Rec and baseline models on amazon datasets.

Dataset	Model	MAE	MSE	RMSE	MAPE
Clothing, Shoes and Jewelry	NeuMF	0.588	0.539	0.734	0.168
	ANR	0.527	0.509	0.713	0.152
	AFRAM	0.500	0.461	0.679	0.145
	MFNR	0.483	0.450	0.670	0.141
	CAMRec	0.397	0.299	0.547	0.110
	MMCN	0.386	0.302	0.549	0.108
	MAS-Rec	0.316	0.257	0.507	0.089
Cell Phones and Accessories	NeuMF	0.595	0.585	0.765	0.182
	ANR	0.544	0.575	0.758	0.168
	AFRAM	0.505	0.529	0.727	0.160
	MFNR	0.500	0.500	0.707	0.156
	CAMRec	0.451	0.403	0.635	0.136
	MMCN	0.444	0.386	0.621	0.126
	MAS-Rec	0.299	0.278	0.527	0.090

Fourth, multimodal recommender systems such as CAMRec and MMCN outperformed review-based models by leveraging the complementary characteristics of textual and visual modalities. Among them, MMCN achieved superior performance, as it captures high-order user-item interactions while mitigating modality heterogeneity through hierarchical contrastive loss. Although CAMRec recorded slightly lower MSE and RMSE scores on average in the Clothing, Shoes and Jewelry category, the performance difference between the

two models was marginal. Notably, unlike MMCN, CAMRec integrates multimodal features at the level of individual interactions, which may allow for more accurate preference prediction in the fashion domain. This can be attributed to the nature of fashion, where clothing is chosen to suit specific contexts such as time, place, and occasion. These contextual nuances make each interaction particularly influential in shaping user preferences.

Finally, the proposed MAS-Rec model achieved the highest performance among all baseline models. This result can be explained as follows. MAS-Rec effectively summarizes user preferences and item features around key aspects by leveraging LLMs, enabling more accurate preference prediction. In addition, visual information is converted into the same representation format as textual reviews through aspect-based summarization using LLaVA, facilitating semantic alignment and interaction between modalities. Furthermore, a co-attention mechanism is applied to the contextual information extracted from both texts and images, allowing the model to selectively integrate complementary information for improved prediction. A fine-tuned BERT model is integrated into MAS-Rec to better identify task-relevant information during aspect-based summarization.

Following the evaluation protocol adopted in prior studies, we conducted statistical significance analysis to validate the robustness of MAS-Rec. Specifically, paired t-tests were performed between MAS-Rec and all baseline models across MAE, MSE, RMSE, and MAPE. Tables 5 and 6 summarize the results, showing that MAS-Rec achieves statistically significant improvements over all baselines across both datasets. These results indicate that the performance gains are consistent and not due to random variation. Furthermore, the improvements remain statistically significant even when compared with strong multimodal models such as CAMRec and MMCN, demonstrating the effectiveness of modeling semantic interactions through LLM-based aspect summarization and co-attention.

TABLE 5. Statistical significance (T-Test) results for MAS-Rec and baseline models on clothing, shoes and jewelry datasets.

Comparison	MAE	MSE	RMSE	MAPE
NeuMF	53.640***	96.504***	82.773***	71.024***
ANR	63.599***	67.491***	62.801***	68.929***
AFRAM	26.995***	43.386***	41.822***	30.528***
MFNR	20.070***	60.924***	54.905***	35.389***
CAMRec	10.432***	5.734**	5.811**	13.380***
MMCN	9.895***	11.128***	11.178***	13.633***

Notes: ***p < 0.001; **p < 0.01; *p < 0.05

B. ABLATION STUDY ON FUSION STRATEGIES IN THE SEMANTIC INTERACTION NETWORK

To address the limitation of conventional fusion approaches, this study adopts a co-attention mechanism to selectively extract important information from LLM-generated aspect-based summarizations. Through this mechanism, semantically related information between user preferences and item

TABLE 6. Statistical significance (T-Test) results for MAS-Rec and baseline models on cell phones and accessories datasets.

Comparison	MAE	MSE	RMSE	MAPE
NeuMF	119.612***	212.175***	153.420***	303.125***
ANR	56.746***	96.064***	84.065***	107.618***
AFRAM	34.457***	145.841***	133.814***	108.361***
MFNR	28.449***	145.351***	125.836***	92.721***
CAMRec	21.876***	13.110***	13.738***	30.377***
MMCN	25.817***	16.911***	17.567***	42.457***

Notes: ***p < 0.001; **p < 0.01; *p < 0.05

characteristics is modeled via mutual referencing, enabling the model to learn more expressive representations. To validate the effectiveness of this approach, we conducted comparative experiments with conventional fusion methods with respect to preference prediction accuracy. Specifically, the performance of Dot, Elementwise, and Concatenation methods was compared, and the experimental results are presented in Table 7.

The Dot method, which relies solely on inner product operations without additional nonlinear transformations, shows the lowest performance, indicating its limitation in capturing complex nonlinear relationships between user preferences and item characteristics. The Elementwise method partially improves performance by modeling feature-wise interactions, but it still suffers from information loss due to its restricted interaction capacity at the element level.

TABLE 7. Performance comparison of fusion strategies in the semantic interaction network of MAS-Rec.

Dataset	Method	MAE	MSE	RMSE	MAPE
Clothing, Shoes and Jewelry	Dot	0.539	0.505	0.711	0.138
	Elementwise	0.355	0.287	0.536	0.098
	Concatenation	0.318	0.264	0.514	0.089
	Co-Attention	0.316	0.257	0.507	0.089
Cell Phones and Accessories	Dot	0.535	0.517	0.719	0.141
	Elementwise	0.376	0.326	0.571	0.107
	Concatenation	0.310	0.289	0.538	0.095
	Co-Attention	0.299	0.278	0.527	0.090

The Concatenation method demonstrates relatively stable performance, as it preserves the full feature representations by directly concatenating user and item features, thereby enabling subsequent layers to learn richer interactions. Notably, the proposed co-attention mechanism achieves the best performance across all evaluation metrics. This is because it effectively models semantic interactions through mutual referencing between user preferences and item characteristics at the aspect level. As a result, the model can focus on more informative and relevant features.

C. PERFORMANCE COMPARISON OF MAS-Rec WITH CONVENTIONAL FEATURE EXTRACTORS

This study aims to improve recommendation accuracy by applying LLMs to multimodal reviews through aspect-based summarization, which more precisely captures the complementary characteristics of texts and images. To verify the effectiveness of this approach, we conducted comparative

experiments using MAS-Rec. The comparison was made against a variant model that adopts conventional methods for extracting modality-specific features in existing multimodal recommender systems. MAS-Rec is a full model that integrates LLMs to capture semantic connections between texts and images and to reflect salient information in preference prediction. In contrast, the variant model removes the LLM components. Instead, it utilizes multimodal features extracted using BERT and ViT, which are commonly employed in prior studies. The experimental results are summarized in TABLE 8.

Based on the experimental results, MAS-Rec consistently outperformed the variant model across all evaluation metrics in both categories. The variant excludes the LLM components and instead adopts conventional methods. This finding suggests that traditional approaches that simply fuse multimodal features are limited in enabling effective cross-modal reference between texts and images [19], [47]. In contrast, applying LLMs enables more accurate modeling of semantic interactions between texts and images, even when the modalities exhibit high levels of heterogeneity. These results support previous findings [51], which suggest that transforming images into textual representations through LLaVA not only unifies expression formats but also preserves rich contextual information, thereby improving recommendation accuracy.

TABLE 8. Performance comparison between MAS-Rec and a conventional approach.

Dataset	Model	MAE	MSE	RMSE	MAPE
Clothing, Shoes and Jewelry	BERT and ViT	0.398	0.315	0.561	0.112
	LLMs	0.316	0.257	0.507	0.089
Cell Phones and Accessories	BERT and ViT	0.447	0.393	0.627	0.133
	LLMs	0.299	0.278	0.527	0.090

D. IMPACT OF PROMPTING STRATEGIES ON LLM-BASED RECOMMENDER SYSTEM

In this study, LLMs are employed to perform aspect-based summarization of multimodal reviews, aiming to accurately capture both user preferences and item characteristics. Based on prior studies, key aspects are predefined, and instruction-based prompts are designed to elicit rich aspect level information that is critical for preference prediction. To evaluate the effectiveness of this approach, we apply instruction prompts commonly used in prior studies to LLMs. Specifically, we analyze the impact of different prompting strategies, including Reasoning, Segment, and the proposed Aspect-based Summarization prompts, on recommendation accuracy. The prompts used in the experiments are illustrated in FIGURE 3 and the corresponding experimental results are reported in TABLE 9.

The Segment prompt instructs LLaMA to describe user characteristics likely to favor a given item, and LLaVA to

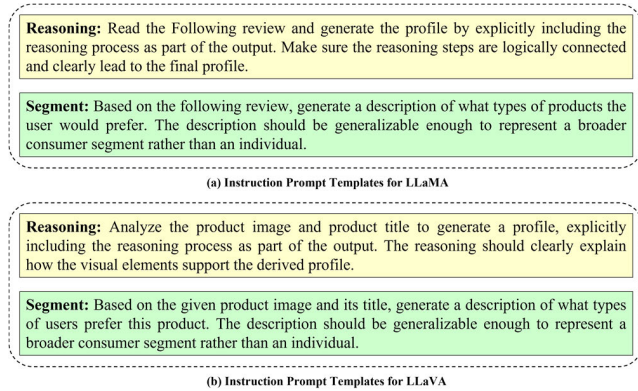


FIGURE 3. Design variants of instruction prompt templates.

describe item characteristics likely to be preferred by a given user [50]. However, as this approach generates explanations based on generalized user segments, it has limitations in capturing individual user preferences. Furthermore, inferring group level characteristics that are not explicitly stated in user reviews can lead to hallucinations, thereby reducing prediction accuracy. As a result, this method demonstrated the lowest performance across all categories.

The Reasoning prompt encourages LLMs to generate reasoning-based responses, enabling a deeper analysis of user preferences and item characteristics [61]. This prompting strategy achieved the best performance in the Clothing, Shoes and Jewelry category. As a domain of experience goods, fashion related items are strongly influenced by users' subjective experiences with individual aspects when evaluating item preferences. This indicates that the Reasoning prompt effectively predicts the aspects users consider important through a process of in-depth reasoning.

TABLE 9. Performance comparison of prompting strategies in MAS-Rec.

Dataset	Model	MAE	MSE	RMSE	MAPE
Clothing, Shoes and Jewelry	Segment	0.342	0.309	0.556	0.098
	Reasoning	0.290	0.229	0.478	0.082
	Aspect-based Summarization	0.316	0.257	0.507	0.089
	Reasoning	0.381	0.392	0.626	0.118
Cell Phones and Accessories	Segment	0.307	0.287	0.535	0.093
	Reasoning	0.299	0.278	0.527	0.090
	Aspect-based Summarization				
	Reasoning				

In contrast, the Reasoning prompt exhibited lower performance in the Cell Phones and Accessories category compared to aspect-based summarization. As a domain of search goods, mobile phones are characterized by clearly stated aspects such as Qualities, Designs, and Durability, where objective features play a substantial role in preference prediction. These results suggest that complex reasoning processes may instead introduce noise and negatively affect predictive accuracy in the mobile phone domain [62].

The proposed Aspect-based Summarization method ranked second across all evaluation metrics in the Clothing,

Shoes and Jewelry category, exhibiting lower performance than the Reasoning prompt. This result suggests that structuring information strictly by aspects may limit the model's ability to fully capture users' affective evaluations of items. In addition, the LLMs employed in this study are not specifically optimized for recommendation tasks, which may constrain their ability to accurately identify nuanced emotional expressions and subtle contextual cues at the aspect level within reviews [48].

In contrast, the Aspect-based Summarization achieved the best performance across all evaluation metrics in the Cell Phones and Accessories category. These results indicate that aspect-based summarization effectively captures user preferences and item characteristics for search goods with clearly defined aspects. In particular, because user evaluations of item aspects in domains of search goods tend to exhibit lower variance, aspect-based summarization is more likely to enhance predictive performance by providing structured and focused representations.

E. ATTENTION ROLLOUT-BASED ANALYSIS OF BERT DECISIONS IN MAS-Rec

In this study, we fine-tuned BERT to better capture preference-relevant information as part of the aspect-based summarization process applied to LLM-generated content. We analyzed the impact of BERT fine-tuning on recommendation performance using the Attention Rollout method. This technique accumulates attention across all hidden layers to more accurately model information flow [63]. It was used to identify which words in the summarized texts were treated as key signals by the BERT encoders. In our analysis, we visualized the tokens receiving the highest attention from the [CLS] token in the fine-tuned BERT model, thereby revealing which information the model considered important for preference prediction.

For this analysis, we applied aspect-based summarization using LLMs to a user's multimodal review of a women's sweater cardigan. The cardigan features a ribbed knit texture in shades of gray, with a scalloped edge along the button placket. To improve visualization clarity, we accounted for BERT's tendency to assign high attention weights to punctuation marks, periods, and special tokens such as [SEP] [64]. These tokens were excluded, and the attention weights were normalized so that their sum equaled one. In addition, token level attention weights were aggregated into word level weights using the mean value, and words within the top 20% of total attention across the input sequence were highlighted in bold. Detailed visualization results are presented in FIGURE 4.

F. DISCUSSION ON HALLUCINATION EFFECTS AND MITIGATION

LLMs, such as LLaMA and LLaVA, provide strong capabilities for generating aspect-based summarization from multimodal reviews. However, since these models are not

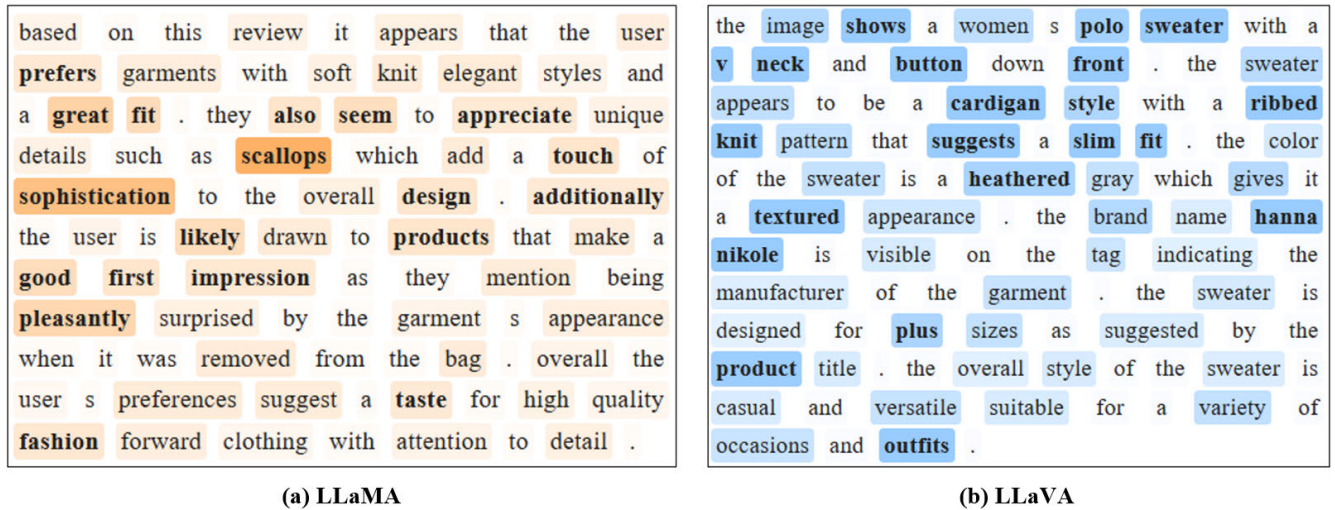


FIGURE 4. Visualization of BERT attention rollout during aspect-based summarization using LLM-generated content.

specifically optimized for recommendation tasks, they may occasionally produce ungrounded information that is not fully supported by the input data. Such ungrounded content typically arises when the model infers user preferences or item characteristics beyond the provided reviews, often relying on generalized reasoning or implicit assumptions rather than input-specific evidence.

To address this issue, our framework leverages aspect-based summarization as a structured representation mechanism that inherently reduces the influence of unreliable LLM outputs. Specifically, this process operates at multiple levels. First, aspect-based summarization constrains the generation space by organizing outputs into predefined aspect categories, thereby reducing the likelihood of producing overly generalized or ungrounded content. Second, the structured nature of aspect-based summarization enables more stable feature extraction in the downstream recommendation model. The fine-tuned BERT encoder captures contextually relevant information while suppressing noisy or less informative signals from the generated outputs. Third, the co-attention mechanism further refines the representations derived from aspect-based summarization by selectively emphasizing semantically aligned interactions between user and item features, effectively attenuating the impact of less reliable signals. Through this process, aspect-based summarization not only provides a unified representation across modalities but also serves as an implicit filtering mechanism that progressively reduces the influence of ungrounded LLM outputs before they contribute to the final prediction.

While hallucination may occur in general LLM-based generation, the observed impact appeared limited within our experimental setting. Although this study did not conduct quantitative evaluation using dedicated hallucination benchmarks, the generated summaries remained relatively stable across repeated experiments and did not result in substantial degradation in recommendation performance. These results

indicate that the impact of potential ungrounded generation is effectively limited in practice. Furthermore, the experimental results demonstrate consistent and robust performance across multiple runs, suggesting that the model is resilient to potential noise introduced by LLM-based summarization. This can be attributed to the structured design of aspect-based summarization and downstream modeling components, which constrain and refine the generated summaries, thereby reducing the propagation of unreliable information.

Overall, these findings suggest that, although hallucination cannot be completely eliminated, its influence on preference prediction is substantially reduced. As a result, the proposed framework achieves robust and reliable recommendation performance despite the inherent limitations of LLM-based generation.

VI. CONCLUSION

Advancements in information and communication technology have accelerated the rapid growth of e-commerce platforms, exposing users to an overwhelming volume of items and service related information. As a result, users face increasing difficulty in selecting items that align with their personal tastes and preferences. To address this challenge, extensive research has been conducted on recommender systems that aim to reduce the time and effort required for information retrieval. Early recommender systems relied solely on user ratings to predict preferences, which limited their ability to reflect users' detailed experiences with items. In response, studies began incorporating review texts to capture more nuanced user feedback; however, most of these approaches did not utilize review images, thereby failing to fully reflect the visual characteristics of items in preference prediction. In contrast, multimodal recommender systems leverage the complementary nature of textual and visual modalities to enhance prediction accuracy. Nevertheless, most existing studies do not account for the structural

and representational heterogeneity between texts and images, often relying on simple feature fusion methods that are insufficient for capturing the semantic interactions between modalities.

To address these limitations, this study proposes MAS-Rec, which utilizes LLMs to transform item related information in review images into textual information, thereby effectively capturing semantic connections across modalities. Specifically, MAS-Rec performs aspect-based summarization on multimodal reviews by applying LLaMA and LLaVA, enabling heterogeneous textual and visual information to be processed within a unified representation framework. In addition, a fine-tuned BERT is incorporated into the preference prediction module to more precisely model the influence of diverse item aspects on user preferences. This framework identifies information that is critical for preference prediction from multimodal reviews, allowing for a more accurate reflection of the value and importance users assign to individual item aspects. Experimental results on datasets collected from Amazon.com show that MAS-Rec outperforms all baseline models across all evaluation metrics. These findings demonstrate that summarizing review images by aspect using LLaVA can significantly enhance recommendation accuracy by reducing modality level inconsistencies.

This study addresses the limitations of existing multimodal recommender systems in effectively handling modality heterogeneity and presents the following academic contributions. First, we demonstrate the effectiveness of utilizing LLaVA to summarize review images into textual descriptions for preference prediction. While prior studies have noted the difficulty in capturing semantic interactions across modalities using traditional multimodal fusion methods, we propose a novel approach that summarizes item characteristics by aspects through LLMs. This approach enables LLMs to process multimodal reviews, despite their structural and representational differences, into a unified textual format, thereby improving the accuracy of preference prediction. Second, we quantitatively show that the strategy used to construct instruction prompts significantly impacts preference prediction performance. We confirmed that aspect-based summarization effectively capture both user preferences and item features in the context of search goods. Third, we verify that a fine-tuned BERT model can accurately extract critical information from summaries generated by LLMs for preference prediction. Moreover, our findings reaffirm the conclusions of prior research that item aspects play a key role in predicting user preferences.

The findings of this study highlight the practical potential of effectively leveraging multimodal data in real-world service environments. Based on this, the following managerial implications can be drawn. First, the proposed model, which predicts user preferences using LLMs, can provide more refined and personalized recommendations. This, in turn, contributes to enhancing user satisfaction and increasing return visit rates by offering an improved platform experience. Second, MAS-Rec is not limited to e-commerce

platforms such as Amazon.com, but can be extended to various domains where multimodal user reviews are available. In particular, it can be effectively applied to domains such as restaurants, hotels, and travel, where visual information plays a significant role in consumer decision-making. In these contexts, the model enables accurate preference prediction by capturing item aspects through aspect-based summarization. Third, unlike prior studies, this approach allows the model to identify context-specific features by summarizing review images according to aspects. When such summaries are provided to users, they can serve as an explanation for the recommendations, thereby supporting more informed purchase decisions.

While this study makes significant contributions to the field of multimodal recommender systems, several limitations must be acknowledged. First, MAS-Rec was evaluated using multimodal reviews from two item categories on Amazon.com. Due to constraints in data collection and computational resources, a smaller dataset was used compared to prior studies in multimodal recommender system research. Therefore, future studies should validate the generalizability of MAS-Rec through experiments on larger and more diverse datasets. Second, aspect-based summarization of multimodal reviews was conducted using LLaMA and LLaVA. However, as these models are not specialized for recommendation tasks, their outputs may occasionally include minor ungrounded content. Nevertheless, the impact of such effects is limited within our framework. Future research could enhance model performance by incorporating user profiles and item descriptions as additional inputs or by adopting low-rank adaptation (LoRA) for task-specific fine-tuning of LLMs. Third, although this study focused on predicting preferences at the level of individual user-item interactions, it is also possible to generate aggregated summaries of users and items based on sets of multimodal reviews. Such summaries could provide richer representations of user preferences and item features, thereby enabling more accurate preference predictions. Lastly, while this study anticipates that leveraging rich multimodal information may help alleviate the data sparsity problem, this hypothesis was not quantitatively verified. Future work should investigate the extent to which LLM-based recommender systems can effectively mitigate data sparsity.

In addition, although the proposed framework demonstrated stable performance across repeated experiments, the factual consistency and reliability of LLM-generated aspect-based summarization were not quantitatively evaluated through dedicated hallucination benchmarks. Since LLMs are not specifically optimized for recommendation tasks, generated summaries may occasionally include generalized or partially ungrounded information. Future studies should further investigate the robustness of multimodal recommender systems under noisy multimodal inputs and incorporate quantitative evaluation methods for factual consistency and reliability. Furthermore, additional research is needed to examine the scalability of the proposed framework

in large-scale recommendation environments and real-time service settings.

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